Status	Finished
Started	Sunday, 25 October 2026, 3:38 PM
Completed	Sunday, 25 October 2026, 3:51 PM
Duration	12 mins 54 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list is very small
- c. When the list contains duplicate elements
- d. When the list is unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The list is sorted
- b. The search continues in the left or right sublist ✓
- c. The search continues from the beginning
- d. The search stops

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. At the beginning of the list
- c. In the entire list
- d. In the right sublist

Question **4**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the element is found in the list ✓
c. When the list contains duplicate elements
d. When the search algorithm finishes

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

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b. The list must be sorted
c. There must be a mechanism to delete and/or insert elements in the list ✓
d. There should be direct access to the middle element in any sublist

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. log n
c. n/2
d. n ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Searching ✓
b. Finding
c. Mining
d. Discovering

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It can only be applied to large lists
- c. It does not work with negative numbers
- d. It is slower than linear search for small lists

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search stops immediately ✓
- b. The search starts over from the beginning
- c. The search backtracks to find duplicate elements
- d. The search continues until the end of the list

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search divides the list into halves.
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search requires the list to be sorted.

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is not the array at all
- b. When the item is the last element in the array
- c. Item is the last element in the array or item is not there at all
- d. When the item is somewhere in the middle of the array ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Hash search
c. Binary search ✓
d. Sequential search

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By using a hash function
b. By dividing the list into halves
c. By sorting the list first
d. By comparing each element in the list sequentially ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is large and unsorted

Question **15**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops
b. The middle element is chosen randomly
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element